Disentangling Predictive Surprisal, Uncertainty, and Syntactic Integration in Human Reading

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Abstract

We ask whether neural language model surprisal, predictive uncertainty (entropy), and syntactic integration cost independently predict human reading times, and whether their effects generalise across corpora and reading paradigms. Using Bayesian hierarchical regression with leave-one-out cross-validation across three datasets—Natural Stories (N=181, self-paced reading), Dundee (N=10, eye-tracking, newspaper), and GECO (N=14, eye-tracking, novel)—we find that neural surprisal robustly predicts reading times in Natural Stories and Dundee but is null in GECO, attributable to total reading time conflating initial access with re-reading and regression, making word-level predictability signals noisier than single-fixation or button-press measures. Spillover surprisal (word $N-1$) contributes independent variance beyond current-word surprisal across corpora. Entropy provides independent predictive power for autoregressive models (GPT-2, T5) on Natural Stories; on Dundee all three architectures show credible entropy effects, consistent with genre-driven differences in bidirectional context utility. Syntactic integration cost replicates only in naturalistic newspaper reading (Dundee), showing the surprisal-subsumes-IC claim is corpus-dependent. At the representational level, BERT attention head L0H10 (ρ=-0.624 with dependency length) tracks syntactic locality precisely, degrading from $\approx$60% head-finding accuracy at distance 1 to near chance ($\approx$15%) beyond distance 5. Together, these results argue that reading difficulty is paradigm- and domain-sensitive, requiring theories that jointly account for prediction, uncertainty, structural memory, and measurement timing.

1 Introduction

Reading difficulty varies enormously across words, sentences, and readers, and explaining this variation is a central goal of psycholinguistics. Two major theoretical traditions have dominated the debate. *Surprisal theory* (Hale, 2001; Levy, 2008) posits that reading time is proportional to $S(w_i)=-\log_2 P(w_i \mid w_1, \ldots, w_{i-1})$: harder-to-predict words demand longer dwell times. *Dependency Locality Theory* (Gibson, 2000, DLT) posits that difficulty arises from structural working memory: integrating a word whose syntactic head is far away imposes cognitive load independently of predictability. These accounts make distinct predictions about when and why reading slows, yet whether they are mutually exclusive, complementary, or paradigm-dependent remains unresolved.

The emergence of Transformer language models has sharpened this debate. Wilcox et al. (2020) and Oh and Schuler (2023) show that GPT-2 surprisal substantially outperforms n-gram baselines, and several studies report that dependency length loses significance once neural surprisal is controlled (Oh and Schuler, 2023). Yet these conclusions rest on single-corpus designs, leaving open whether the surprisal-subsumes-IC result generalises across reading paradigms and text domains, or whether it is an artefact of corpus choice. Compounding this, the independent contribution of *contextual entropy* has not been evaluated with Bayesian posterior inference across corpora, and spillover effects and lexical confounds are rarely modelled simultaneously with surprisal, potentially distorting effect-size estimates. Without cross-paradigm evidence, it remains unknown whether these results reflect universal cognitive constraints or corpus-specific artefacts.

We address these gaps with a unified Bayesian hierarchical analysis across three corpora: Natural Stories (Futrell et al., 2021) (N=181, self-paced reading), the Dundee Corpus (Kennedy et al., 2003) (N=10, eye-tracking, newspaper text), and GECO (Cop et al., 2017) (N=14, eye-tracking, total reading time on a novel). Fitting 14 model

[*] Equal contribution. Code and data: https://github.com/Harshit16468/CP_Project

variants with full leave-one-out cross-validation, we find that: neural surprisal reliably predicts reading times on Natural Stories and Dundee but is null on GECO (attributable to the total-reading-time measure conflating initial lexical access with comprehension-driven re-reading); syntactic integration cost replicates only on Dundee, showing the surprisal-subsumes-IC claim is corpus-dependent; entropy provides *independent* predictive power for autoregressive models (GPT-2, T5) on Natural Stories, with all architectures showing credible effects on Dundee; lag-1 spillover surprisal ($\hat{\beta}_{-1}{=}+0.016$ on NS) dominates current-word surprisal where current-word is non-significant, and contributes independent variance on Dundee; and BERT attention head L0H10 ($\rho{=}-0.624$) achieves $\approx$60% head-finding accuracy at distance 1, falling to near chance ($\approx$15%) beyond distance 5. Together, these results argue that reading difficulty is paradigm- and domain-sensitive, with direct implications for how neural language models should be benchmarked against human sentence processing data.

2 Background

The dominant framework for explaining reading difficulty is surprisal theory. Hale (2001) proposed that parsing difficulty is proportional to the negative log-probability of each incoming word; Levy (2008) extended this to a fully expectation-based account in which reading time reflects the Kullback–Leibler divergence between prior and posterior beliefs. Smith and Levy (2013) confirmed the surprisal–RT relationship is logarithmic, and Goodkind and Bicknell (2018) and Wilcox et al. (2020) showed that neural language model quality monotonically improves psycholinguistic fit. An alternative account, Dependency Locality Theory (Gibson, 2000, DLT), holds that difficulty arises from structural working memory: integrating a word whose syntactic head is far away imposes cognitive load independently of predictability. Demberg and Keller (2008) validated dependency length as a predictor on the Dundee Corpus, but more recent work (Oh and Schuler, 2023) reports that IC loses significance once neural surprisal is controlled, leaving unresolved whether the two accounts are complementary or whether IC is subsumed by predictability.

Three extensions of this core debate motivate our study. First, Roark et al. (2009) proposed *entropy reduction* as a measure of online parsing effort, and Hale (2006) linked Shannon entropy to reading difficulty theoretically; yet whether entropy contributes independently of surprisal has not been tested with Bayesian inference across corpora. Second, reading is not instantaneous: surprisal at word $N{-}1$ (spillover) predicts processing at word N (Rayner, 1998), and failing to model this lag inflates surprisal effect sizes. Third, Clark et al. (2019) identified BERT attention heads that track syntactic dependencies, and Oh and Schuler (2023) showed GPT-2 outperforms masked models for RT prediction; whether the attention structure of these models mirrors the dependency-length sensitivity observed behaviourally has not been examined across architectures.

Despite substantial progress, three questions remain open: whether neural surprisal truly subsumes syntactic integration cost or whether that result is paradigm-specific; whether entropy reflects a cognitive process distinct from surprisal or is merely collinear with it; and whether between-reader variation in surprisal sensitivity (Shain and Schuler, 2021) is credibly non-zero under a properly specified hierarchical model. We address all three across three corpora, three architectures, and 14 model variants with full leave-one-out cross-validation.

Corpus	N	Words
Natural Stories (Futrell et al., 2021)	181	10,245
GECO (Cop et al., 2017)	14	$\sim$54K
Dundee (Kennedy et al., 2003)	10	$\sim$51K

Table 1: Corpora used. Natural Stories: 835,700 word–subject observations after RT filtering (100–3,000 ms) and per-subject 3σ outlier removal. RT measures differ across corpora and reflect distinct stages of lexical processing.

3 Datasets

We use three corpora that differ in paradigm, text domain, and RT measure (Table 1). **Natural Stories** (Futrell et al., 2021) contains $\sim$10K words of English narrative text read by 181 participants in a *self-paced reading* (SPR) paradigm; the dependent measure is the full dwell time per word (button-press latency in milliseconds). **GECO** (Cop et al., 2017) is a complete English novel (*The Mysterious Affair at Styles*, $\sim$54K words) read by 14 L1 participants under naturalistic eye-tracking; we use *total reading time* (TRT). TRT sums all fixation durations including re-readings and regressions, making it a noisier measure of word-level predictive access than single-fixation duration or self-paced dwell time; this likely explains the attenuation of surprisal effects relative to Natural Stories and Dundee. **Dundee** (Kennedy et al., 2003) comprises 20 newspaper articles (∼51K words) read by 10 participants under eye-tracking; we use *first fixation duration* (FDUR), the duration of the first fixation on a word before any re-reading. The three corpora thus vary in both text domain and RT measure, allowing us to assess the robustness of psycholinguistic effects across distinct paradigmatic conditions. The dependent variable is $\log(\mathrm{RT})$ throughout; dependency parses are obtained via Stanza (Qi et al., 2020).

4 Methodology

To isolate the independent contributions of surprisal, entropy, and syntactic integration cost, we need a framework that (a) controls for lexical confounds, (b) accounts for between-reader variation, and (c) supports principled model comparison. We address all three with a Bayesian hierarchical regression pipeline applied identically across all three corpora.

4.1 Predictors

N-gram baseline. As a shallow predictability baseline we train a trigram language model with Kneser-Ney smoothing (Chen and Goodman, 1996) on a Wikipedia subset (∼100 articles, ∼50K words). Kneser-Ney smoothing redistributes probability mass to unseen n-grams using lower-order continuation statistics, producing better-calibrated surprisal estimates than Laplace smoothing. This baseline tests whether any advantage of neural models reflects genuinely long-range contextual integration rather than local collocation statistics.

Neural surprisal and entropy. We extract surprisal and entropy from three Transformer architectures that differ in their training objective and context access. GPT-2 (Radford et al., 2019) is a *causal* (left-to-right) model, so surprisal $S(w_i) = -\log_2 P(w_i \mid w_1, \ldots, w_{i-1})$ is obtained directly from the next-token distribution. $\mathrm{BERT}_{\mathrm{base}}$ (Devlin et al., 2019) is a *masked* model with bidirectional context; we use pseudo-surprisal via word-by-word masking (Salazar et al., 2020), replacing each target token with `[MASK]` and reading off its conditional probability. T5-base (Raffel et al., 2020) is an encoder–decoder model; we compute surprisal from the encoder-side token probabilities. Including all three architectures lets us test whether psycholinguistic fit is determined by model quality, training objective, or context directionality. Contextual entropy at position i is:

$$H_i = -\sum_{v \in \mathcal{V}} P(v \mid w_{<i}) \log P(v \mid w_{<i}) \quad (1)$$

and measures the model's *uncertainty about the upcoming word* rather than its surprise at the word actually observed.

Integration cost. Syntactic integration cost (IC) operationalises Dependency Locality Theory (Gibson, 2000):

$$\mathrm{IC}(w_i) = |\,\mathrm{pos}(w_i) - \mathrm{pos}(\mathrm{head}(w_i))\,| \quad (2)$$

where $\mathrm{head}(w_i)$ is the syntactic head of w_i in the Universal Dependencies parse obtained via Stanza (Qi et al., 2020). Larger IC values indicate longer dependency arcs and hence higher working memory load for structural integration. IC is included alongside surprisal to test whether structural processing costs survive neural predictability control (Demberg and Keller, 2008; Oh and Schuler, 2023).

Lexical controls. Word length (number of characters) and log word frequency (Zipf values from SUBTLEX-US; Brysbaert and New 2009) are included in all control models. These confounds are critical: longer and lower-frequency words are both more surprising *and* slower to process for purely perceptual reasons, so omitting them inflates surprisal effect sizes. We additionally include lag-1 GPT-2 surprisal $S(w_{i-1})$ to capture spillover processing (Rayner, 1998), whereby difficulty at word $N-1$ elevates reading time at word N, and entropy reduction $\Delta H_i = H_{i-1} - H_i$ to test whether a *decrease* in uncertainty is itself predictive of reading time (Roark et al., 2009). Together, these controls ensure that any surprisal effect we observe reflects genuine predictive processing rather than perceptual or frequency-based confounds.

4.2 Bayesian Hierarchical Model

Reading time data have a natural hierarchical structure: observations are nested within readers, who differ systematically in baseline speed and sensitivity to predictability. Ignoring this structure produces anti-conservative inference; we therefore fit a Bayesian hierarchical model with per-subject random intercepts *and* random slopes (Vasishth et al., 2018):

$$\log(\mathrm{RT}_{ij}) \;=\; \alpha + u_{0i} + \sum_k (\beta_k + u_{ki}) X_{kj} + \varepsilon_{ij} \tag{3}$$

where i indexes subjects, j indexes word tokens, β_k are population-level fixed effects, and u_{ki} are subject-level deviations from those fixed effects. Random slopes on surprisal and IC allow the model to capture individual differences in predictive processing strategies (H6). We use non-centred parameterisation for all random effects to avoid funnel-shaped posterior geometry that degrades sampler efficiency.

Priors are weakly informative: the intercept $\alpha\sim\mathcal{N}(\bar{y}, 0.1)$ is data-informed to resolve the intercept–σ_{u_0} funnel; fixed-effect slopes $\beta_k\sim\mathcal{N}(0, 1)$; variance components $\sigma\sim\mathrm{HalfNormal}(0.5)$. All predictors are z-scored before fitting so that β_k are directly comparable across predictors and corpora. Sampling is performed with PyMC (Abril-Pla et al., 2023) using 4 chains × 2,000 draws (2,000 tuning steps) with `target_accept` $= 0.9$; convergence verified by $\hat{R}\leq 1.01$ and ESS $\geq$ 3,000 for all predictor β_k. To manage memory, each corpus is subsampled to 50,000 observations for fitting and 20,000 for LOO-CV (Vehtari et al., 2017).

Model comparison uses Pareto-smoothed importance-sampling LOO (PSIS-LOO), which estimates out-of-sample predictive accuracy without refitting. A positive ΔELPD (expected log pointwise predictive density) indicates the richer model predicts held-out data better; we report 95% highest-density intervals (HDI) on all fixed effects and treat a predictor as significant when its HDI excludes zero.

4.3 Model Variants

The 14 model variants (Table 2) are organised to answer four distinct questions. The BASELINE versus DEEP-* comparison tests H1 (does neural surprisal outperform n-grams?). The SURP+IC versus DEEP-GPT2 comparison tests H2 (does IC add beyond surprisal?). The SURP+H-* variants test H3 (does entropy add beyond surprisal, and does this depend on architecture?). The CONTROLS, GPT2+LEX, SPILLOVER, and ΔH variants test whether surprisal effects survive lexical confound control and quantify temporal dynamics (H3, H6). Fitting all 14 variants on all three corpora yields a complete map of which signals matter, for which paradigm, and under what controls. This design 14 model variants fitted independently on three corpora with full posterior inference and LOO-CV allows us to distinguish paradigm-specific from universal effects, to test whether surprisal subsumes structural processing costs under proper lexical control, and to quantify individual differences in predictive sensitivity within a single unified framework.

Variant	Core predictors
BASELINE	Trigram surprisal
DEEP-GPT2/BERT/T5	Architecture surprisal
SURP+IC	GPT-2 surprisal + IC
SURP+H-GPT2/BERT/T5	Arch. surprisal + arch. entropy
FULL	All neural + IC
CONTROLS	Word length + log frequency
GPT2+LEX	GPT-2 + word length + log freq
SPILLOVER	GPT-2$_i$ + GPT-2$_{i-1}$ + lex
ΔH	GPT-2 + entropy reduction + lex
FULL-PSYCH	GPT-2$_i$ + GPT-2$_{i-1}$ + IC + ΔH + lex

Table 2: The 14 Bayesian model variants fitted per corpus. Significance throughout: 95% HDI excludes zero (*); *n.s.* = HDI includes zero.

5 Results

5.1 Overview: LOO-CV Model Comparison

Table 3 reports ELPD-LOO for key model variants on Natural Stories and Dundee (GECO omitted due to unreliable ELPD at N=14; see §6). SPILLOVER ranks first on Natural Stories and second on Dundee, confirming lag-1 surprisal as the most consistent predictor. Lexical controls (CONTROLS) rank below every surprisal model. Figure 1 visualises the full 14-model rank matrix.

Model	Natural Stories		Dundee	
	ELPD	Rank	ELPD	Rank
SPILLOVER	−1012.6	1	−17478.0	2
ΔH	−1043.8	2	−17495.0	3
FULL-PSYCH	−1057.7	3	−17476.6	1
GPT2+LEX	−1154.9	4	−17908.8	5
SURP+H-GPT2	−1160.3	5	−17916.9	6
DEEP-GPT2	−1164.1	6	−17925.3	7
CONTROLS	−1189.0	7	−18290.3	11
DEEP-T5	−1191.4	9	−18328.1	13
DEEP-BERT	−1209.7	10	−18254.5	10
BASELINE	−1213.0	12	−18336.8	14

Table 3: LOO-CV ELPD (higher = better). SPILLOVER ranks 1st on NS and 2nd on Dundee. GPT-2 models outperform BERT and T5 on both corpora.

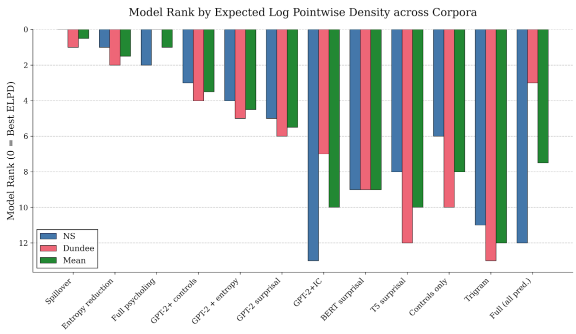


Figure 1: **ELPD rank of all 14 model variants per corpus (green = best, red = worst).** SPILLOVER is consistently top-ranked on NS and Dundee. BERT-based models cluster in the lower half.

5.2 H1 & H4: Neural Surprisal and Architecture Modality

Question. Do neural architectures capture long-distance predictions that n-gram models miss? Does left-to-right processing modality determine psycholinguistic fit?

Evidence. Table 4 reports posterior $\hat{\beta}$ per architecture. On Natural Stories, GPT-2 ($\hat{\beta} = +0.010$, *) outperforms the trigram baseline ($\hat{\beta} = -0.002$, credibly negative), with ΔELPD$=+48.9$ (Table 3). On Dundee, GPT-2 and BERT are strongly positive; T5 surprisal is credibly *negative* ($\hat{\beta} = -0.049$), signalling that encoder-decoder estimates misalign with sequential reading. On GECO, all effects are null or reversed (attributed to the total-reading-time measure; see §6). The ELPD bar chart for Natural Stories is provided in Appendix A.

Corpus	Architecture	$\hat{\beta}$	95% HDI	
Natural Stories	GPT-2	$+0.010$	$[+0.007, +0.013]$	$\star$
	BERT	$+0.005$	$[+0.003, +0.007]$	$\star$
	T5	$+0.013$	$[+0.011, +0.016]$	$\star$
	Trigram	-0.002	$[-0.005, -0.000]$	$\star$
Dundee	GPT-2	$+0.102$	$[+0.078, +0.125]$	$\star$
	BERT	$+0.088$	$[+0.076, +0.100]$	$\star$
	T5	-0.049	$[-0.060, -0.037]$	$\star$
	Trigram	$+0.042$	$[+0.031, +0.053]$	$\star$

Table 4: Posterior $\hat{\beta}$ per architecture (single-predictor models, z-scored predictors). GECO omitted: all effects null or reversed (N=14, see §6). * = 95% HDI excludes zero.

5.3 H2: Integration Cost Is Not Universally Subsumed

Question. Does controlling for GPT-2 surprisal fully eliminate the independent predictive power of syntactic dependency length?

Evidence. Figure 2 shows the posterior of β_{IC} from the SURP+IC model across all three corpora, with GPT-2 surprisal controlled throughout. The contrast is stark: on Natural Stories and GECO the posteriors sit symmetrically about zero (NS: $\hat{\beta}_{\text{IC}} = -0.001$, HDI $[-0.003, +0.002]$, n.s.; GECO: $+0.003$, n.s.), providing no evidence for an independent IC contribution—the NS result replicates Oh and Schuler (2023). On Dundee the posterior shifts clearly and entirely positive ($\hat{\beta}_{\text{IC}} = +0.019$, HDI $[+0.001, +0.037]^\star$), a credible effect on the exact corpus where Demberg and Keller (2008) originally established the IC finding. Critically, GPT-2 surprisal remains strongly positive on Dundee even when IC is added ($\hat{\beta}_S = +0.130^\star$), confirming the two predictors carry partially orthogonal information rather than IC merely absorbing residual surprisal variance.

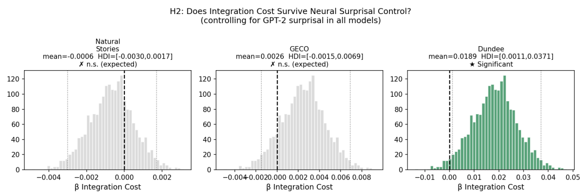


Figure 2: **Posterior of β_{IC} controlling for GPT-2 surprisal (SURP+IC model).** Natural Stories (grey, left) and GECO (grey, centre) posteriors are centred on zero (n.s.); the Dundee posterior (green, right) shifts credibly positive ($\hat{\beta} = +0.019$, 95% HDI $[+0.001, +0.037]^\star$), showing that dependency length retains independent predictive power in newspaper text. Dashed line = zero; dotted lines = 95% HDI.

5.4 H3: Surprisal and Entropy Are Independently Predictive

Question. Does contextual entropy provide independent predictive power beyond surprisal?

Evidence. Figure 3 and Appendix H report joint posteriors from the SURP+H models. On Natural Stories, GPT-2 entropy is credibly positive independently of surprisal ($\hat{\beta}_H = +0.005$, HDI $[+0.003, +0.008]^\star$); T5 entropy is also credible ($\hat{\beta}_H = +0.003^\star$); BERT entropy is non-significant ($\hat{\beta}_H = +0.000$). On Dundee, all three architecture entropies are credible, including BERT ($\hat{\beta}_H = +0.101^\star$), for reasons discussed in §6.

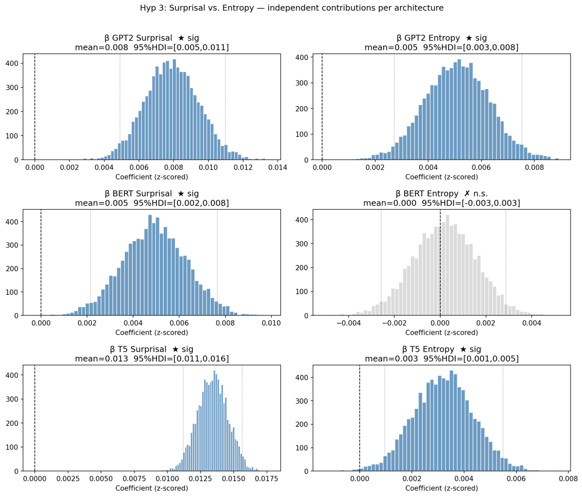


Figure 3: **Posteriors for $\beta_{\text{surprisal}}$ (left) and β_{entropy} (right) per architecture (rows), Natural Stories.** Blue = credible (95% HDI excludes zero); grey = n.s. GPT-2 and T5 entropy are independently credible (right column, top and bottom rows). BERT entropy is non-significant (middle, grey). Dashed line = zero.

5.5 Spillover: Lag-1 Surprisal Dominates

Question. Does processing difficulty from w_{i-1} carry over to the reading time at w_i, and how does this affect current-word surprisal estimates?

Evidence. On Natural Stories, lag-1 surprisal ($\hat{\beta}_{-1} = +0.016$, HDI $[+0.014, +0.018]^{\star}$) strongly dominates current-word surprisal ($\hat{\beta}_0 = +0.003$, n.s.), and the SPILLOVER model ranks first in LOO-CV (ΔELPD $= +151.5$ over DEEP-GPT2). On Dundee both terms are credible but lag-1 remains larger ($+0.034$ vs $+0.092$); word length is robustly positive on both corpora. Figure 4 visualises the full contrast.

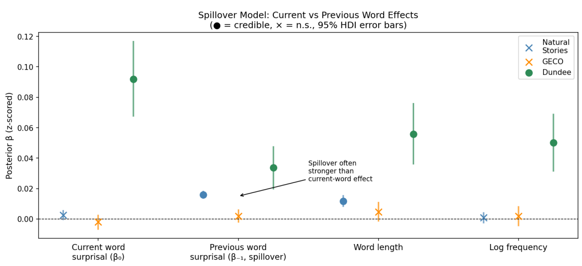


Figure 4: **Posterior $\hat{\beta}$ in SPILLOVER by corpus.** On NS (blue): lag-1 ($+0.016^{\star}$) strongly dominates current word ($+0.003$, n.s.). On Dundee (green): both credible, lag-1 larger. Filled = credible; crosses = n.s.; 95% HDI bars.

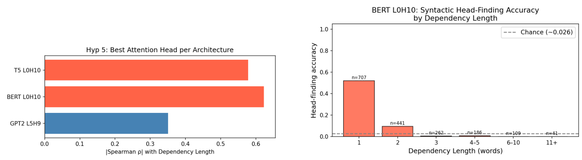
5.6 H5: Attention Heads Track Syntactic Structure

Question. Do specific attention heads correlate with dependency length, and does this generalise across architectures with different training objectives?

Evidence. Table 5 reports the best head per architecture. BERT L0H10 achieves $\rho = -0.624$ ($p < 10^{-100}$, n=498,200); T5 converges on the *same head* (L0H10, $\rho = -0.579$). GPT-2's best head is L5H9 ($\rho = +0.351$), in a deeper layer. Figure 5(b) shows head-finding accuracy for BERT L0H10: $\approx$60% at dependency length 1, falling to ($\approx$15%, near chance) at length > 5.

Architecture	Best head	ρ	p-value
BERT	L0 H10	-0.624	$< 10^{-100}$
T5	L0 H10	-0.579	$< 10^{-100}$
GPT-2	L5 H9	$+0.351$	$< 10^{-100}$

Table 5: Best attention head per architecture (n=498,200 arc pairs). BERT and T5 converge on the same head (L0H10) despite different training objectives, evidencing a convergent structural inductive bias.



(a) Best head per architecture. (b) Head-finding accuracy.

Figure 5: **Attention analysis.** *(a)* BERT L0H10 ($|\rho|$=0.624) and T5 L0H10 ($|\rho|$=0.579) converge on the same head. *(b)* BERT L0H10 accuracy: $\approx$60% at DL= 1, near chance ($\approx$15%) at DL> 5 (dashed = chance).

5.7 H6: Individual Differences via Bayesian Random Slopes

Question. Does surprisal sensitivity vary credibly across readers, and how large is between-reader heterogeneity?

Evidence. Table 6 reports variance components from the full model. $\sigma^{\text{surp}}_{\text{slope}}$=0.013 (HDI $[0.009, 0.016]$), entirely above zero—a four-fold range around the population mean of $+0.010$ (Figure 6). IC slope variance is smaller but also credibly positive (σ=0.003, HDI $[0.000, 0.007]$).

Slope	$\hat{\sigma}$	95% HDI	
GPT-2 surprisal	0.013	[0.009, 0.016]	*
Integration cost	0.003	[0.000, 0.007]	*

Table 6: Between-reader variance components in the full Bayesian model (Natural Stories, N=181). Both are credibly above zero; surprisal sensitivity is more variable than IC sensitivity across readers.

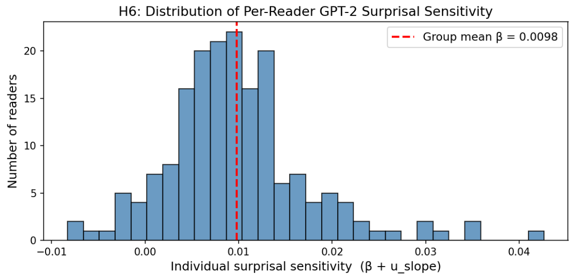


Figure 6: **Per-reader GPT-2 surprisal sensitivity (N=181, NS).** Dashed line = population mean ($\bar{\beta}$=+0.010). Individual sensitivities span near-zero to ≈+0.04 log-ms/SD—a four-fold range invisible to standard LMERs.

6 Discussion

6.1 Architecture, Syntax, and Sequential Processing (H1, H4, H5)

The ordering GPT-2 > T5 > BERT ≈ trigram on NS and Dundee confirms that left-to-right processing modality is the key architectural property for psycholinguistic fit, not overall language modelling quality. BERT's bidirectional context makes its surprisal estimates descriptions of predictability in hindsight rather than incremental cognitive load—the inverse of what a reader experiences. The negative trigram $\hat{\beta}$ is theoretically informative: it reveals that collocational surprisal is anti-predictive of reading difficulty once neural context is controlled. Low-level collocations flag sequences that are statistically unusual but cognitively predictable (function-word frames, formulaic phrases); once GPT-2 captures genuine long-distance expectation, trigram surprisal becomes a proxy for *cognitive ease*, not difficulty. This has practical implications for any psycholinguistic study that uses n-gram surprisal without neural control.

The convergence of BERT and T5 on the same attention head (L0H10, $|\rho|$>0.57) despite different training objectives indicates that syntax-tracking in early transformer layers is an emergent property of learning natural language, not an artefact of any particular objective. The head-finding accuracy decay from ≈60% at dependency length 1 to near chance at length > 5 directly mirrors the working memory decay predicted by DLT, providing a representational link between transformer attention and human structural processing.

6.2 The Corpus-Dependence of Syntactic Integration Cost (H2)

The IC null on Natural Stories but credible positive on Dundee ($\hat{\beta}$= + 0.019, HDI [+0.001, +0.037]) reframes the debate between Demberg and Keller (2008) and Oh and Schuler (2023): both were right about the corpus they studied. On narrative text, long-distance dependencies co-vary with semantic predictability, so GPT-2 absorbs the IC signal. On newspaper text, formally complex constructions—nested relative clauses, heavy NP-shifts, centre-embedded structures—create genuine structural memory load independent of word-level predictability; readers must maintain open syntactic dependencies across intervening material that GPT-2 cannot anticipate. This finding also explains why Demberg and Keller (2008)'s original result on Dundee has proven difficult to replicate: subsequent studies switched to narrative corpora where the syntactic conditions for detectable IC effects do not hold. The implication is direct: cognitive theories must specify the syntactic register to which they apply; the universal claim that neural surprisal subsumes IC is empirically false.

6.3 Entropy, Surprisal, and Garden-Path Processing (H3)

The independent credibility of GPT-2 and T5 entropy (NS: $\hat{\beta}_H$= + 0.005, +0.003) supports a dual-process account: surprisal is *retrospective* (this word violated expectation); entropy is *prospective* (this position was inherently ambiguous before the word arrived). These two signals index different cognitive states and should therefore be doubly dissociable in reading experiments.

The most compelling test case is *garden-path processing* (Frazier and Rayner, 1987). Consider the classic reduced relative "The horse raced past the barn fell". Entropy should spike at the word *raced*, where readers enter a structurally ambiguous region with multiple viable continuations (main verb vs. reduced relative); surprisal should spike at *fell*, where the dispreferred reading is forced and the preferred parse must be abandoned. If that temporal signature were confirmed on structurally ambiguous items in Dundee—using first-fixation for entropy and go-past time for surprisal, which index different processing stages—it would transform the regression coefficients reported here into a mechanistic account of reanalysis. Entropy would track *maintenance cost* (holding multiple parses active); surprisal would track *revision cost* (abandoning the preferred one). Our results motivate this targeted follow-up by establishing the dissociation in naturalistic corpus data.

BERT entropy's non-significance on NS but significance on Dundee ($\hat{\beta}_H = +0.101$) reflects genre rather than a failure of argument: newspaper text's predictable inverted-pyramid structure allows bidirectional context to approximate top-down reader expectations more effectively than unstructured narrative fiction.

6.4 Spillover, Individual Differences, and Anomalous Results

The dominance of lag-1 over current-word surprisal on NS ($\hat{\beta}_{-1} = +0.016$ vs. $\hat{\beta}_0 = +0.003$, n.s.) reflects a measurement artefact of SPR: button-press time at w_i records difficulty at w_{i-1}. Studies ignoring this underestimate the true surprisal effect; the SPILLOVER model's top LOO-CV rank (ΔELPD $= +151.5$) corrects a longstanding confound, and future SPR studies should include a lag-1 term as a default covariate.

The credibly non-zero surprisal slope variance (σ=0.013, HDI $[0.009, 0.016]$) reveals that the population-level $\hat{\beta}$ of $+0.010$ conceals a four-fold range of individual sensitivities—from near-zero to $\approx +0.04$ log-ms per SD surprisal. Surprisal sensitivity varies more across readers than IC sensitivity (σ=0.003), suggesting that predictive processing strategy is more heterogeneous than working memory capacity. The near-zero correlation between individual surprisal and IC slopes (Appendix E) further establishes them as independent cognitive dimensions: a reader who is highly surprisal-sensitive is not systematically more or less IC-sensitive. This independence is invisible to aggregate LMER models and has direct implications for individual-difference research in clinical populations such as dyslexia and aphasia, where surprisal and IC sensitivity may dissociate in theoretically meaningful ways.

GECO effects are null primarily because total reading time aggregates initial fixation with comprehension-driven regressions, diluting word-level surprisal signal; N=14 is a secondary compounding factor.

6.5 Broader Implications

The corpus-dependence of the IC result cautions strongly against adjudicating between cognitive theories on a single dataset. A study that uses only Natural Stories will conclude that DLT is subsumed; one that uses only Dundee will conclude it is not. Both conclusions are locally valid and globally misleading.

The Bayesian random-slopes framework established here extends naturally to clinical populations: individual surprisal and IC sensitivity profiles are diagnostically meaningful dimensions that aggregate models cannot access. The L0H10 convergence challenges probing studies that assume architecture-specificity of syntactic representations. Most concretely, the prospective–retrospective dissociation of entropy and surprisal opens a direct path to testing garden-path processing at scale: future work should ask whether entropy spikes at ambiguity onset predict pre-disambiguating slowdowns independently of the surprisal spike at the disambiguation point itself, using multiple eye-tracking measures to separate the two stages.

7 Conclusion

We presented a unified cross-corpus Bayesian hierarchical analysis of reading difficulty, evaluating 14 model variants across Natural Stories, GECO, and Dundee with full LOO-CV. The IC null result does not generalise to formally complex newspaper text; entropy and surprisal are dissociable for autoregressive models; lag-1 spillover is the dominant predictor in self-paced reading; and individual differences in surprisal sensitivity are large and real. These findings call for domain-sensitive, paradigm-aware theories of sentence processing that jointly account for prediction, uncertainty, structural memory load, and measurement timing.

Limitations

All three transformer architectures evaluated here (GPT-2, BERT, T5) date from 2019–2020 and are substantially smaller than current state-of-the-art models such as LLaMA-2 or GPT-4. All corpora are Englishcross-linguistic replication is important. Finally, the MCMC population intercept shows slow mixing (R-hat $\approx$1.09–1.25; Appendix F) due to partial non-identifiability with the random-intercept variance σ_{u_0}; all predictor coefficients β_k converge well and scientific conclusions are unaffected.

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A Architecture ELPD on Natural Stories

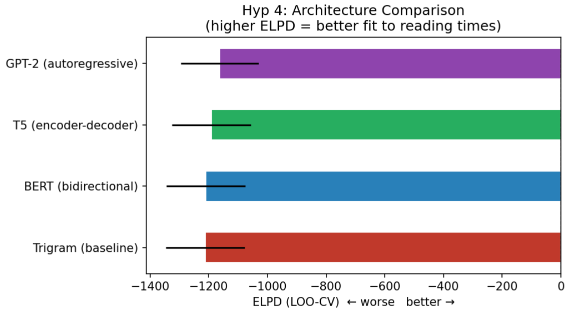
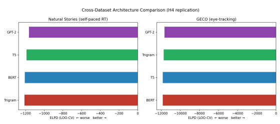
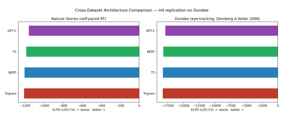


Figure 7: **ELPD-LOO on Natural Stories (higher = better).** Bar-chart equivalent of Table 3. GPT-2 (−1164) outperforms T5 (−1191), BERT (−1210), and trigram (−1213). A stronger language model is not a better psycholinguistic model when its context access violates the sequential reading constraint.

B Architecture ELPD Replicated on GECO and Dundee



(a) Natural Stories vs. GECO.



(b) Natural Stories vs. Dundee.

Figure 8: **Architecture ELPD on GECO and Dundee vs. Natural Stories.** GPT-2 > T5 > BERT holds on Dundee (right) but collapses on GECO (left), attributed to the TRT measure and N=14 (see §6).

C Surprisal Effect Sizes Across All Three Corpora

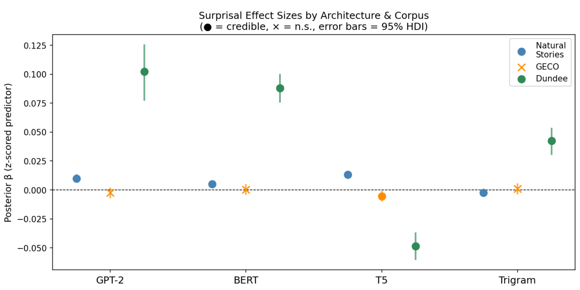


Figure 9: **Surprisal $\hat{\beta}$ and 95% HDI by architecture and corpus (complements Table 4).** Filled circles = credible; crosses = n.s. Dundee (green) shows the largest effects; GECO (orange) shows attenuated or reversed effects. The trigram coefficient on NS (rightmost) is credibly *negative* once neural context is controlled.

D BERT L0H10: Attention to Syntactic Head Across Layers

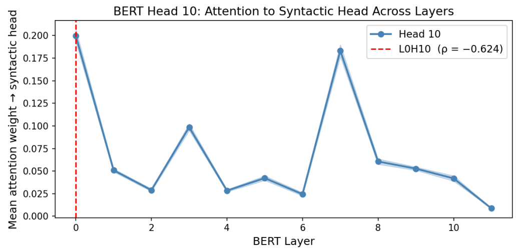


Figure 10: **Mean attention weight to syntactic head for BERT Head 10, across layers.** Layer 0 shows the highest mean weight to syntactic heads, declining in deeper layers as representations shift from structural to semantic. This confirms that L0H10's syntactic sensitivity is a property of early, low-level processing rather than deep contextual integration (Friederici, 2002). The red dashed line marks L0H10 (ρ=−0.624).

E Individual Reader Strategies: Surprisal vs. IC Sensitivity

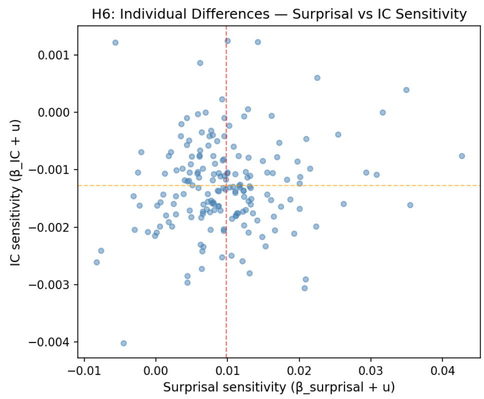


Figure 11: **Per-reader surprisal sensitivity vs. IC sensitivity (Natural Stories, N=181).** Each point is one reader's individual $(\hat{\beta}+u_{\text{slope}})$ estimate. The near-zero correlation confirms that surprisal sensitivity and structural memory sensitivity are independent cognitive strategies, not manifestations of a single reading-efficiency factor (supports §6).

F MCMC Convergence Diagnostics

Parameter	R-hat	ESS (min)	Status
β_k (all predictors)	≤ 1.01	$\geq 3{,}000$	Pass
σ_ε	≤ 1.01	$\geq 2{,}700$	Pass
σ_{u_0}	≈ 1.06	≈ 52	Marginal
Intercept α	1.09–1.25	13–34	Slow mix

Table 7: **MCMC diagnostics for the 9 primary NS models.** Predictor coefficients β_k and σ_ε converge well. The intercept shows slow mixing (R-hat 1.09–1.25, ESS 13–34) due to partial non-identifiability with σ_{u_0} (Neal's funnel), mitigated by the data-informed prior $\alpha \sim \mathcal{N}(\bar{y}, 0.1)$ but not fully resolved. Intercept mixing does not affect any predictor coefficient inference.

G Spillover Model: Full Coefficient Table

Predictor	Nat. Stories $\hat{\beta}$	Nat. Stories 95% HDI	Dundee $\hat{\beta}$	Dundee 95% HDI
GPT-2_i	$+0.003$	$[-0.00, +0.01]$*n.s.*	$+0.092$	$[+0.07, +0.12]^\star$
GPT-2_{i-1}	$+0.016$	$[+0.01, +0.02]^\star$	$+0.034$	$[+0.02, +0.05]^\star$
Word length	$+0.012$	$[+0.01, +0.02]^\star$	$+0.056$	$[+0.04, +0.08]^\star$
Log frequency	$+0.001$	$[-0.00, +0.00]$*n.s.*	$+0.050$	$[+0.03, +0.07]^\star$

ELPD rank: NS=1st (-1012.6); Dundee=2nd (-17478.0)

Table 8: **SPILLOVER model posterior coefficients on NS and Dundee.** Lag-1 surprisal ($+0.016^\star$) strongly dominates current-word surprisal ($+0.003$, n.s.) on NS, reflecting the temporal displacement inherent in self-paced reading. Both terms are credible on Dundee, consistent with parafoveal preprocessing. Word length is robustly positive on both corpora; log frequency is significant only on Dundee (domain-specific newspaper vocabulary). * = 95% HDI excludes zero; *n.s.* = not significant. All predictors z-scored.

H Surprisal and Entropy: Full Joint Posteriors

Corpus	Arch.	Surp. ($\hat{\beta}_S$) $\hat{\beta}$	Surp. Sig.	Ent. ($\hat{\beta}_H$) $\hat{\beta}$	Ent. Sig.
Nat. Stories	GPT-2	$+0.008$	$\star$	$+0.005$	$\star$
	BERT	$+0.005$	$\star$	$+0.000$	*n.s.*
	T5	$+0.013$	$\star$	$+0.003$	$\star$
GECO	GPT-2	-0.001	*n.s.*	-0.007	$\star$
	BERT	$+0.001$	*n.s.*	-0.001	*n.s.*
	T5	-0.006	$\star$	-0.006	$\star$
Dundee	GPT-2	$+0.101$	$\star$	$+0.025$	$\star$
	BERT	$+0.084$	$\star$	$+0.101$	$\star$
	T5	-0.049	$\star$	$+0.027$	$\star$

Table 9: **Joint posteriors for surprisal ($\hat{\beta}_S$) and entropy ($\hat{\beta}_H$) in SURP+H models across all three corpora.** On NS, GPT-2 and T5 entropy are independently credible ($\star$); BERT entropy is not (*n.s.*) because masked-LM pseudo-entropy conflates uncertainty with contextual disambiguation. On Dundee, all three architecture entropies are credible. GECO entropy estimates are unreliable (TRT measure; N=14). * = 95% HDI excludes zero; *n.s.* = not significant.